

ST MARY 1 SSW, MT, USA

WMO: 742290

Lat: 48.7412N

Lon: 113.4330W

Elev: 1388

StdP: 85.72

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 04130

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.8	-24.3	-31.6	0.2	-27.8	-28.3	0.3	-24.6	9.7	0.6	8.9	-0.1	1.0	N/A	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	28.0	13.5	26.5	13.1	24.8	12.8	15.3	23.7	14.4	23.5	13.5	22.8	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.3	17.2	10.7	9.5	16.9	9.6	8.8	16.5	47.5	23.7	44.7	23.3	41.9	23.0	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.8	7.0	-33.4	31.3	2.8	2.2	-35.4	32.9	-37.1	34.1	-38.7	35.3	-40.7	36.9		
(5)			-34.2	17.0	2.6	1.3	-36.0	18.0	-37.5	18.7	-39.0	19.4	-40.8	20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-4.4	-5.6	-1.1	2.8	7.6	11.4	16.0	14.9	10.5	5.1	-0.4	-4.9		
	DBStd	9.31	7.98	7.92	6.22	4.58	3.99	3.22	2.81	3.13	4.32	5.27	6.90	7.69		
	HDD10.0	2556	447	436	345	219	97	22	1	4	46	164	313	462		
	HDD18.3	5111	704	669	603	467	334	210	83	111	237	411	561	721		
	CDD10.0	501	0	0	0	2	22	62	186	156	60	12	2	0		
	CDD18.3	15	0	0	0	0	0	1	9	4	1	0	0	0		
	CDH23.3	671	0	0	0	0	4	38	335	241	51	1	0	0		
	CDH26.7	127	0	0	0	0	0	5	72	45	4	0	0	0		
(14) Wind	WSAvg	3.1	4.1	3.4	3.4	3.0	2.6	2.7	2.4	2.2	2.6	3.4	4.0	3.9		
(15) Precipitation	PrecAvg	568	39	33	37	45	70	96	42	38	42	41	47	39		
	PrecMax	833	90	70	83	83	143	201	164	87	95	100	146	104		
	PrecMin	361	6	4	15	11	14	30	2	0	2	2	3	8		
	PrecStd	128	21	18	20	17	38	55	41	25	25	25	37	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	9.9	13.0	18.5	23.7	27.1	31.2	30.2	27.3	21.5	15.4	10.4		
		MCWB	2.9	3.6	5.0	7.4	10.8	13.5	15.0	13.7	12.5	9.9	6.8	3.6		
	2%	DB	8.7	8.0	10.2	15.3	20.9	24.0	28.7	27.8	24.8	17.4	12.7	7.8		
		MCWB	2.6	2.9	3.7	6.3	10.0	13.6	13.6	13.3	11.9	8.5	5.6	2.0		
	5%	DB	6.5	6.2	8.4	12.8	18.9	21.6	26.9	26.2	22.6	15.2	10.0	6.0		
		MCWB	1.6	2.1	3.1	5.4	9.4	12.0	13.6	12.8	11.2	7.2	4.9	1.6		
	10%	DB	4.7	4.2	6.8	10.5	16.3	19.4	25.2	24.2	19.9	13.3	7.9	4.1		
		MCWB	0.6	0.8	2.3	4.3	8.3	10.9	13.2	12.5	10.3	6.4	3.3	0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.2	5.6	8.8	11.7	15.7	17.6	15.7	14.4	10.8	7.7	4.6		
		MCDB	9.6	8.7	10.9	17.7	21.5	21.8	24.9	23.4	21.6	17.8	14.0	9.1		
	2%	WB	2.9	3.3	4.5	6.9	10.6	13.9	15.6	14.8	12.9	9.2	6.8	2.9		
		MCDB	6.9	7.2	9.0	14.4	19.4	22.2	24.2	24.1	21.7	16.0	12.6	6.0		
	5%	WB	1.7	2.3	3.4	5.5	9.8	12.4	14.8	13.9	11.8	7.9	5.2	1.7		
		MCDB	5.2	6.1	7.7	11.7	18.0	20.1	23.7	23.9	21.2	13.8	9.8	5.1		
	10%	WB	0.8	0.5	2.3	4.3	8.8	11.3	13.9	13.1	10.7	6.7	3.6	0.6		
		MCDB	4.1	3.8	6.3	9.5	16.5	17.8	23.1	22.7	19.6	12.6	7.4	3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	8.6	9.1	10.7	11.1	14.8	14.9	12.1	8.6	7.1	7.3		
		MCDBR	8.9	9.4	10.7	14.5	17.2	17.5	19.8	20.7	19.7	13.2	8.8	7.7		
		MCWBR	6.1	4.4	6.6	7.9	8.5	8.3	8.7	8.6	8.8	6.9	5.0	4.5		
		MCDBR	8.8	7.9	9.2	13.1	15.5	15.7	17.7	17.9	17.1	11.7	8.4	6.7		
	5% WB	MCWBR	6.7	5.5	5.8	7.3	7.7	8.1	8.5	8.0	8.1	6.5	5.1	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.238	0.259	0.294	0.304	0.308	0.308	0.318	0.295	0.276	0.246	0.228		
	taud		2.401	2.402	2.432	2.405	2.505	2.547	2.560	2.514	2.554	2.578	2.526	2.406		
	Ebn at Noon		869	944	971	959	958	952	947	924	919	888	849	826		
	Edh at Noon		63	80	93	107	101	98	95	95	82	68	56	54		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.23	3.43	4.70	5.60	6.28	7.14	5.77	4.10	2.49	1.36	0.99		
	RadStd		0.10	0.17	0.32	0.30	0.51	0.52	0.44	0.39	0.34	0.23	0.10	0.08		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page